

Mini E-Commerce Backend

Products Module Blueprint

1. Products Module Overview

Responsibilities:

- Create Product
- Get All Products
- Get Product By ID
- Update Product
- Delete Product
- Search Products

2. Products APIs

Create Product

Item	Value
Method	POST
Endpoint	/products
Authorization	Admin
Response	201 Created

Example Request

```
{
  "title": "iPhone 16",
  "description": "Apple Phone",
  "price": 1200,
  "category": "Mobiles",
  "stock": 25,
  "image": "image.jpg"
}
```

Get All Products

Item	Value
Method	GET
Endpoint	/products
Authorization	Public
Response	200 OK

Get Product By ID

Item	Value
Method	GET
Endpoint	/products/:id

Authorization	Public
Response	200 OK

Update Product

Item	Value
Method	PATCH
Endpoint	/products/:id
Authorization	Admin
Response	200 OK

Delete Product

Item	Value
Method	DELETE
Endpoint	/products/:id
Authorization	Admin
Response	200 OK

3. Search Products

GET /products?search=iphone
GET /products?category=mobiles

Purpose:

- Search by title
- Search by category

4. Product Request Flow

Client → Request → JWT Middleware → Admin Middleware → Product Controller → MongoDB → Response

5. Authorization Matrix

Operation	User	Admin
Browse Products	Yes	Yes
Get Product	Yes	Yes
Create Product	No	Yes
Update Product	No	Yes
Delete Product	No	Yes

6. Folder Structure

```
controllers/  
  product.controller.js
```

```
routes/  
  product.route.js
```

7. Controller Responsibilities

Function	Responsibility
createProduct	Create Product
getProducts	Get All Products
getProduct	Get Product By ID
updateProduct	Update Product
deleteProduct	Delete Product

8. HTTP Status Codes

Code	Description
200	Success
201	Product Created
400	Validation Error
401	Unauthorized
403	Forbidden
404	Product Not Found
500	Internal Server Error

9. Common Error Scenarios

Error	Status Code
Product Not Found	404
Unauthorized User	401
Forbidden Access	403
Validation Error	400

10. Testing Checklist

- Create Product
- Get All Products
- Get Product By ID
- Update Product
- Delete Product

- Search Products
- Authorization
- Validation
- Error Handling

11. Interview Notes

Why PATCH instead of PUT?

Because PATCH updates only the required fields, while PUT replaces the entire resource.

Why can only Admin create products?

To prevent normal users from modifying the product catalog.

What is the difference between Browse and Search?

- Browse returns all products.
- Search filters products based on user input.

When do we return 401?

When the user is not authenticated or the JWT token is invalid.

When do we return 403?

When the user is authenticated but lacks permission.

Why do we return 201?

Because a new resource has been successfully created.